

Characterization of Galaxies via Optical and Infrared Surveys

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May 31, 2026

Abstract

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1 Introduction

With ongoing and upcoming surveys providing ever larger amounts of data, the task of object classification is becoming increasingly relevant. In this work, we will summarize and compare select methods for characterizing narrow emission line galaxies and the extraction of their physical properties. For all steps in the following sections, we use data from the latest SDSS data release and match those to spectra obtained by WISE for the same objects (Wright et al. 2010, Kollmeier et al. 2026). Information to reproduce these results are included in the appendix.

2 Optical & Infrared Classification

Generally, the broad labeling of galaxies as Active Galactic Nuclei (AGN) or Star Formation (SF) dominated types relies on reconstructing the excitation and ionization mechanisms from emission lines, and determining whether these match the thermal continuum of stellar populations or the strongly nonthermal radiation produced in AGN powered by Super Massive Black Holes (SMBH) in their centers. Approaches analyzing the continuum emission provide complementary insights, and by probing different wavelengths we can peer through obscuring dust and gas to unveil the true underlying physics. Now, we will explore some famous examples.

2.1 BPT Diagram

By plotting the ratio of emission line fluxes for [OIII] at 5007 Å to Hβ at 4863 Å versus [NII] at 6583 Å to Hα at 6565 Å in their work, Baldwin, Phillips, and Terlevich draw a diagram that sorts galaxies into a distinctive winged shape (Baldwin et al. 1981). Due to lacking the energies to ionize heavier atoms, SF types tend towards the lower left of the image, while powerful AGN fall into the upper right tip. By choosing emission lines close to one another, this framework automatically cancels out reddening differences as well as calibration errors. Depicted in Figure 1 is our entire dataset as a scatter plot that has been shaded based on the density

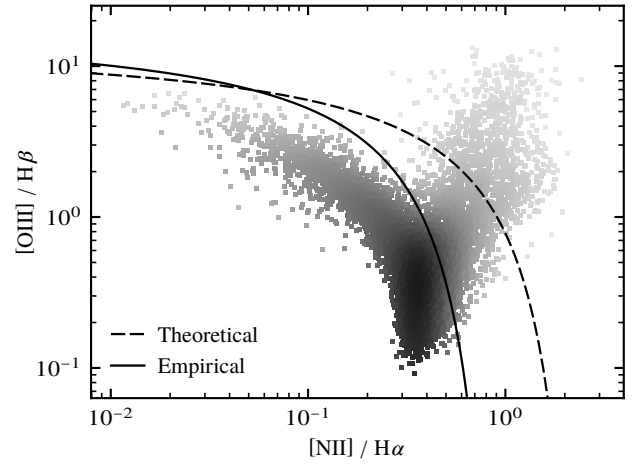


Figure 1: Standard BPT diagram and class separation curves (Kewley et al. 2001, Kauffmann et al. 2003).

of nearby points. This convention is adopted for all such diagrams in this work for better interpretability of the data. Together with this, standard theoretical (Kewley et al. 2001) and empirical (Kauffmann et al. 2003) curves to discriminate between AGN and SF classes are shown.

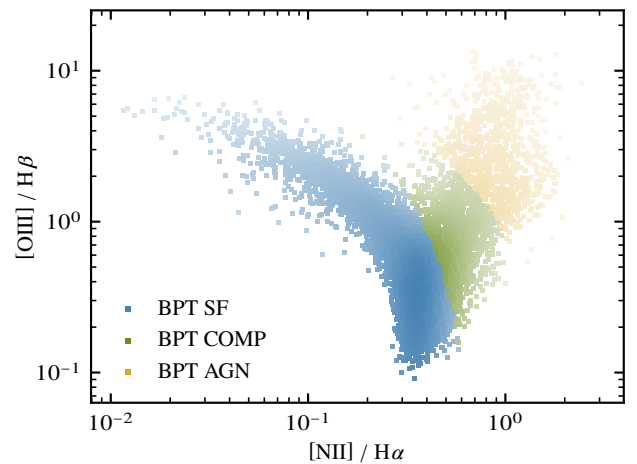


Figure 2: Colored BPT diagram with derived class labels.

We label points between the lines as COMP for composite types and display the classes in Figure 2 encoded by color. Additionally, we have access to the already processed SDSS labels which we can similarly plot in the BPT diagram. This is done in Figure 3 for the AGN and SF as well as starburst SB subclasses. Comparing to the previous plot gives a first visual impression of their cross method agreement, which for the AGN wing seems acceptable, but will be evaluated more rigorously in the coming paragraphs.

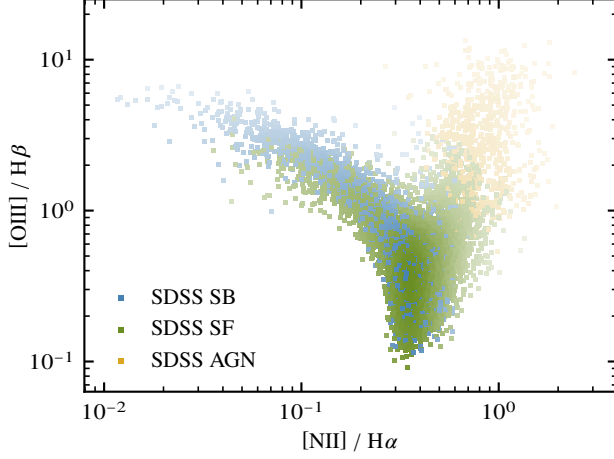


Figure 3

2.2 WHAN Diagram

Next up is the WHAN diagram, which plots the equivalent width $W_{H\alpha}$ of the H α line against the ratio of [NII] to H α fluxes. Since equivalent width measures line strength in terms of the underlying continuum, it is also independent of measurement effects.

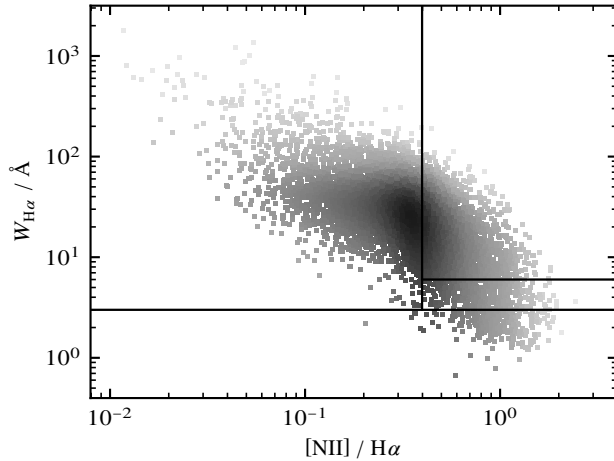


Figure 4

2.3 Category Mixing

2.4 WISE Colors

2.5 Evaluating Diagnostics

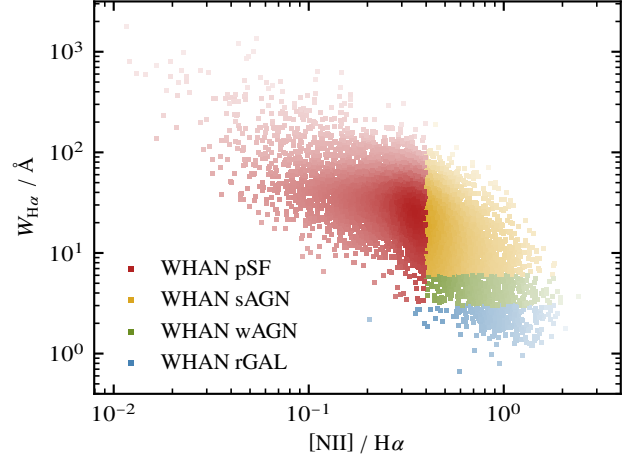


Figure 5

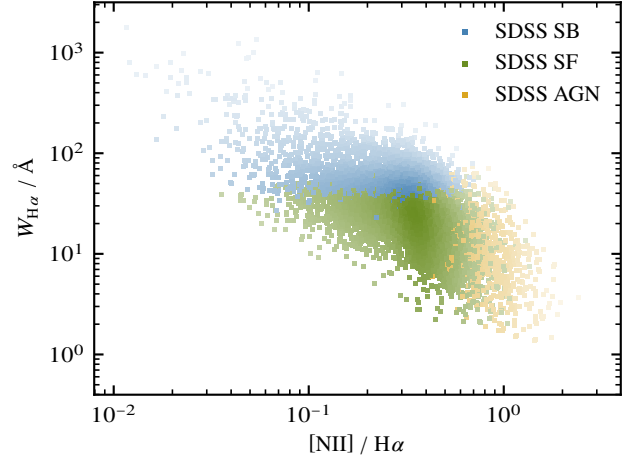


Figure 6

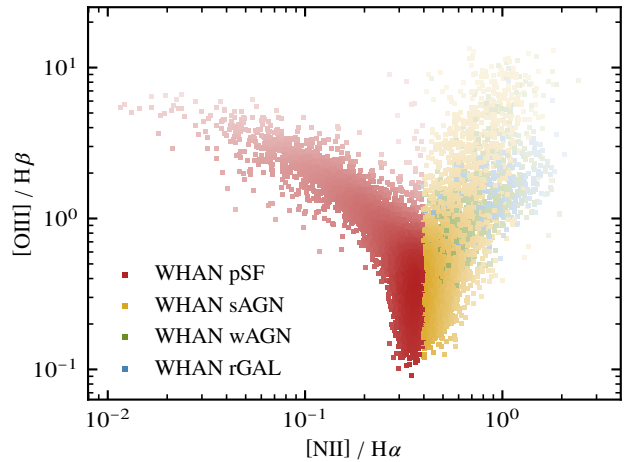


Figure 7

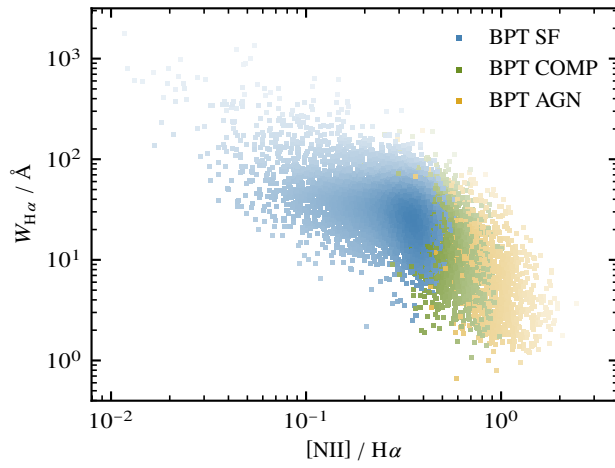


Figure 8

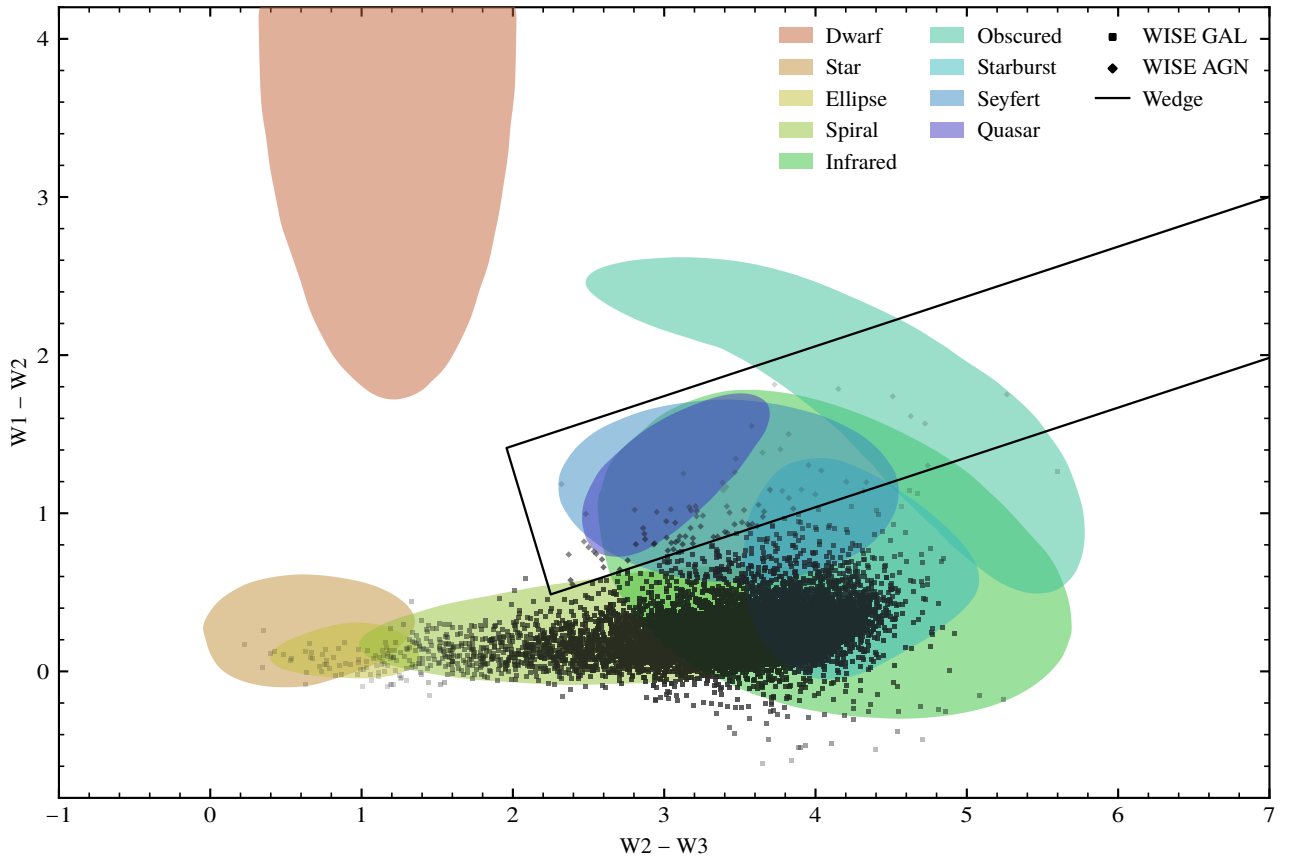


Figure 9

SDSS	1.000	0.680	0.301	0.145
BPT	0.680	1.000	0.271	0.144
WHAN	0.301	0.271	1.000	0.062
WISE	0.145	0.144	0.062	1.000
	SDSS	BPT	WHAN	WISE

Figure 10

SDSS	1.000	0.433	0.301	0.145
BPT	0.433	1.000	0.631	0.104
WHAN	0.301	0.631	1.000	0.062
WISE	0.145	0.104	0.062	1.000
	SDSS	BPT	WHAN	WISE

Figure 11

3 Physical Properties

3.1 Gas Density

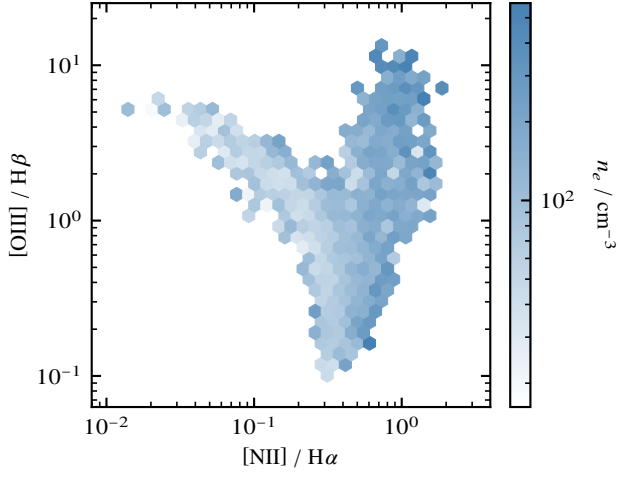


Figure 12

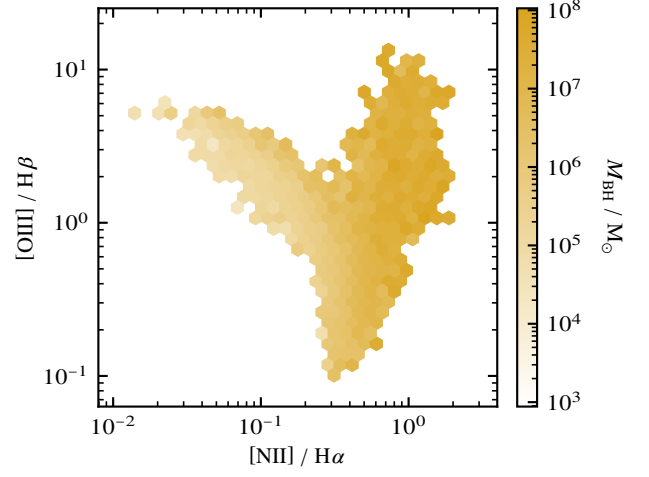


Figure 14

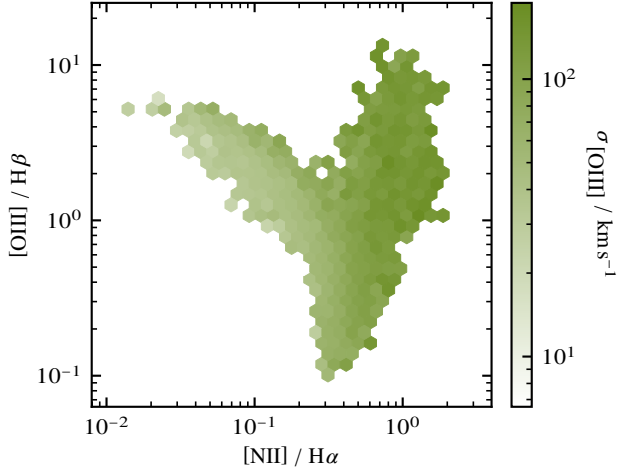


Figure 13

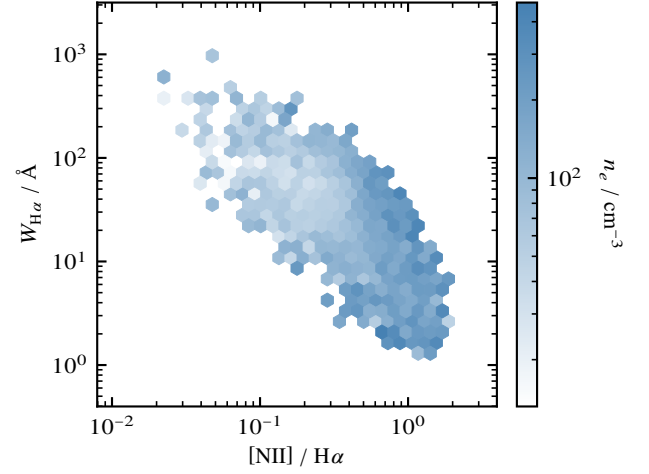


Figure 15

3.2 Gas Velocity

3.3 Black Hole Mass

3.4 Data Analysis

3.5 Conclusion

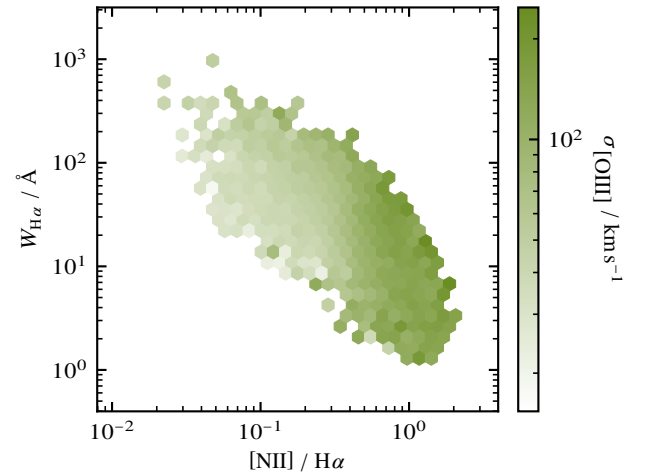


Figure 16

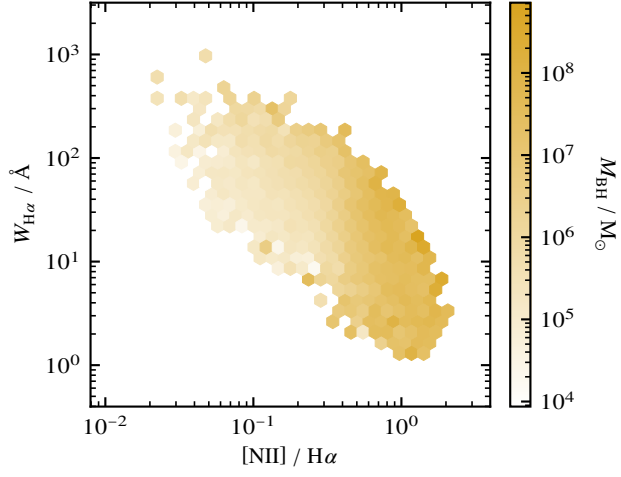


Figure 17

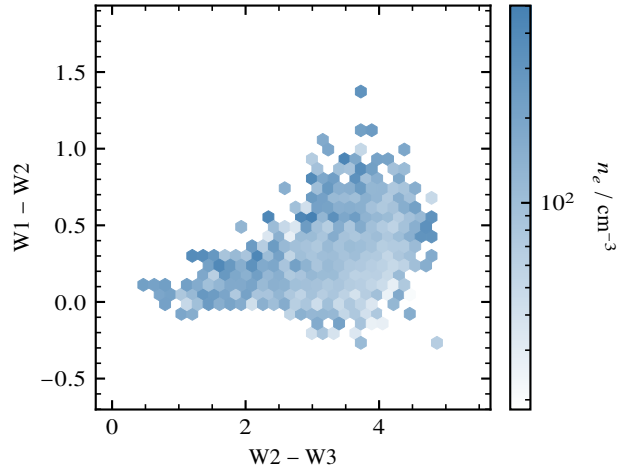


Figure 18

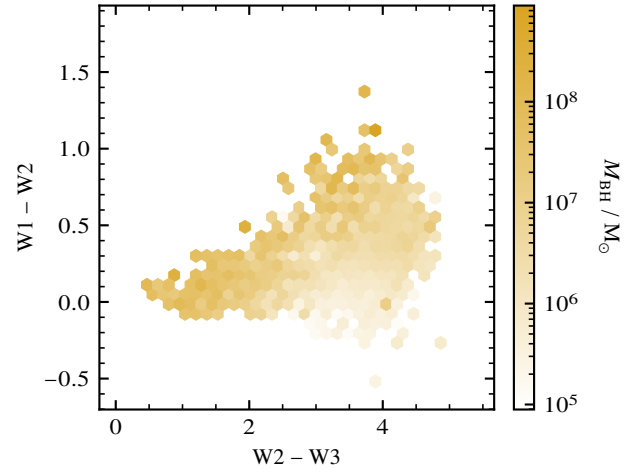


Figure 20

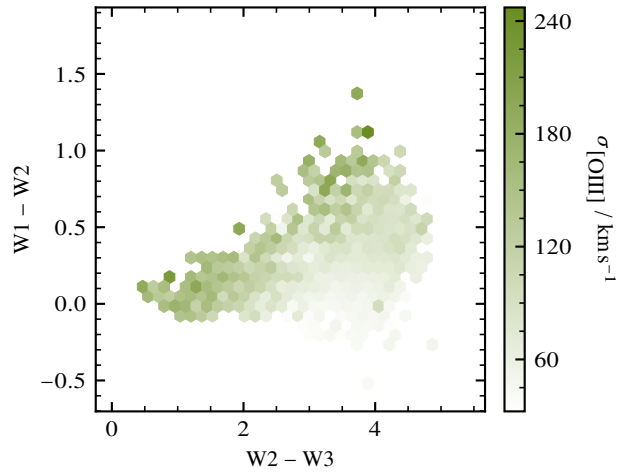


Figure 19

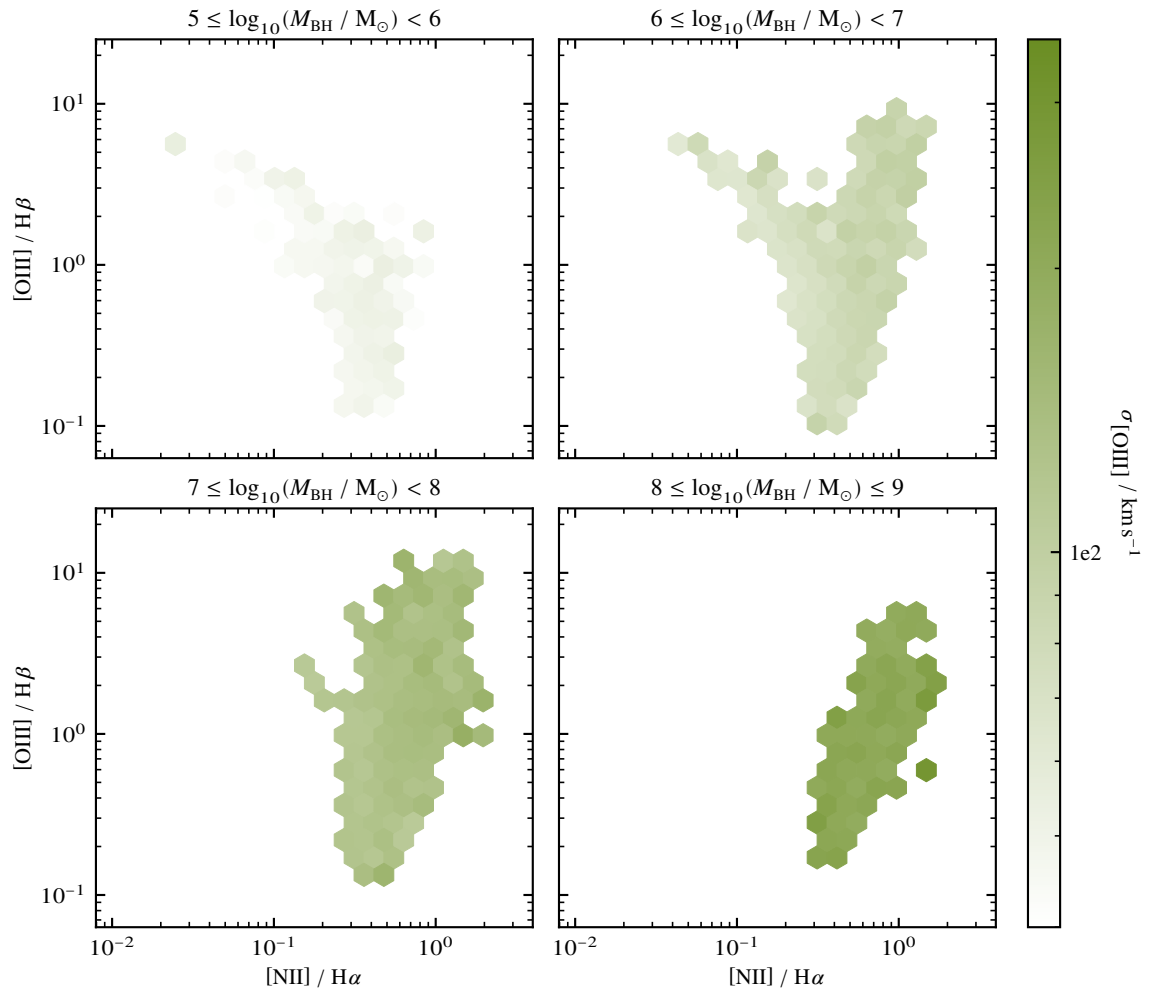


Figure 21

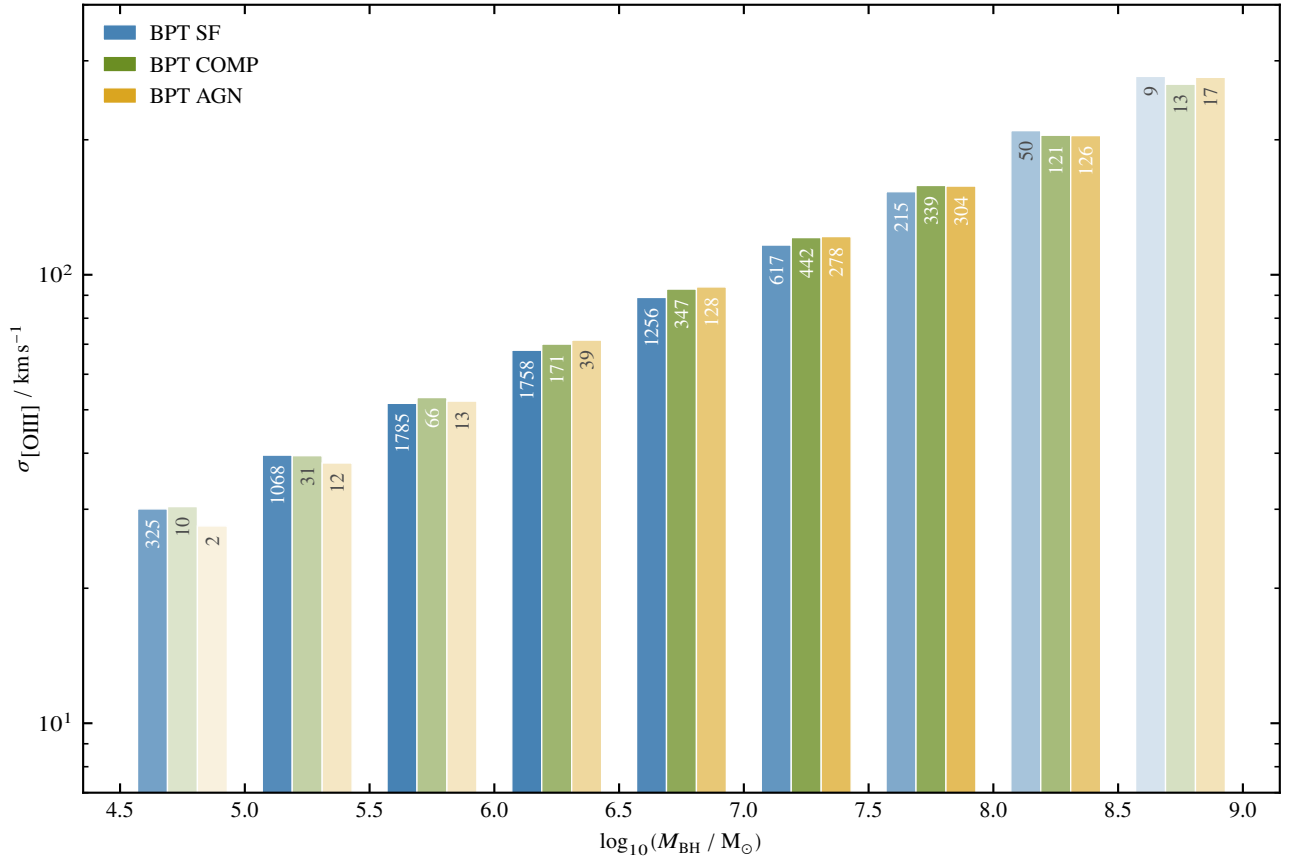


Figure 22

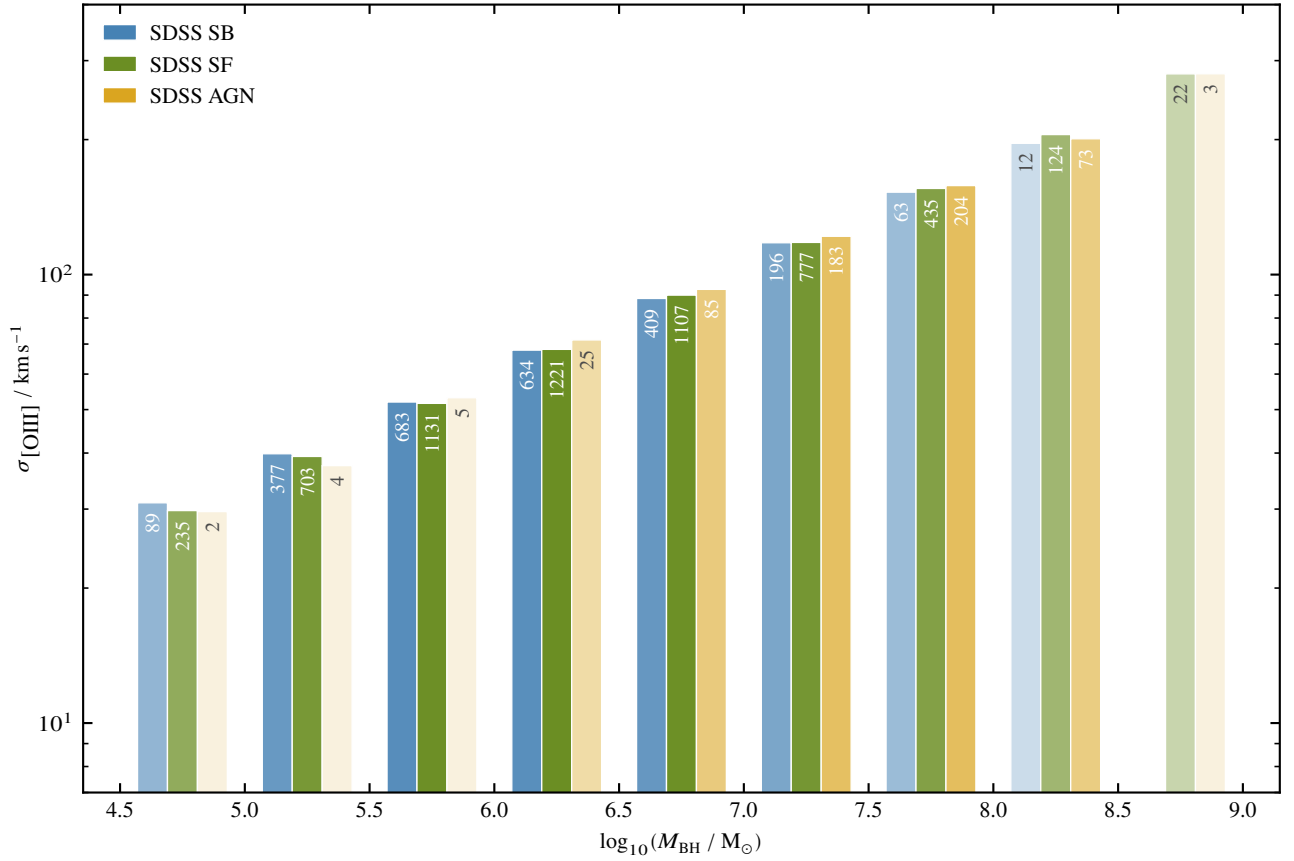


Figure 23

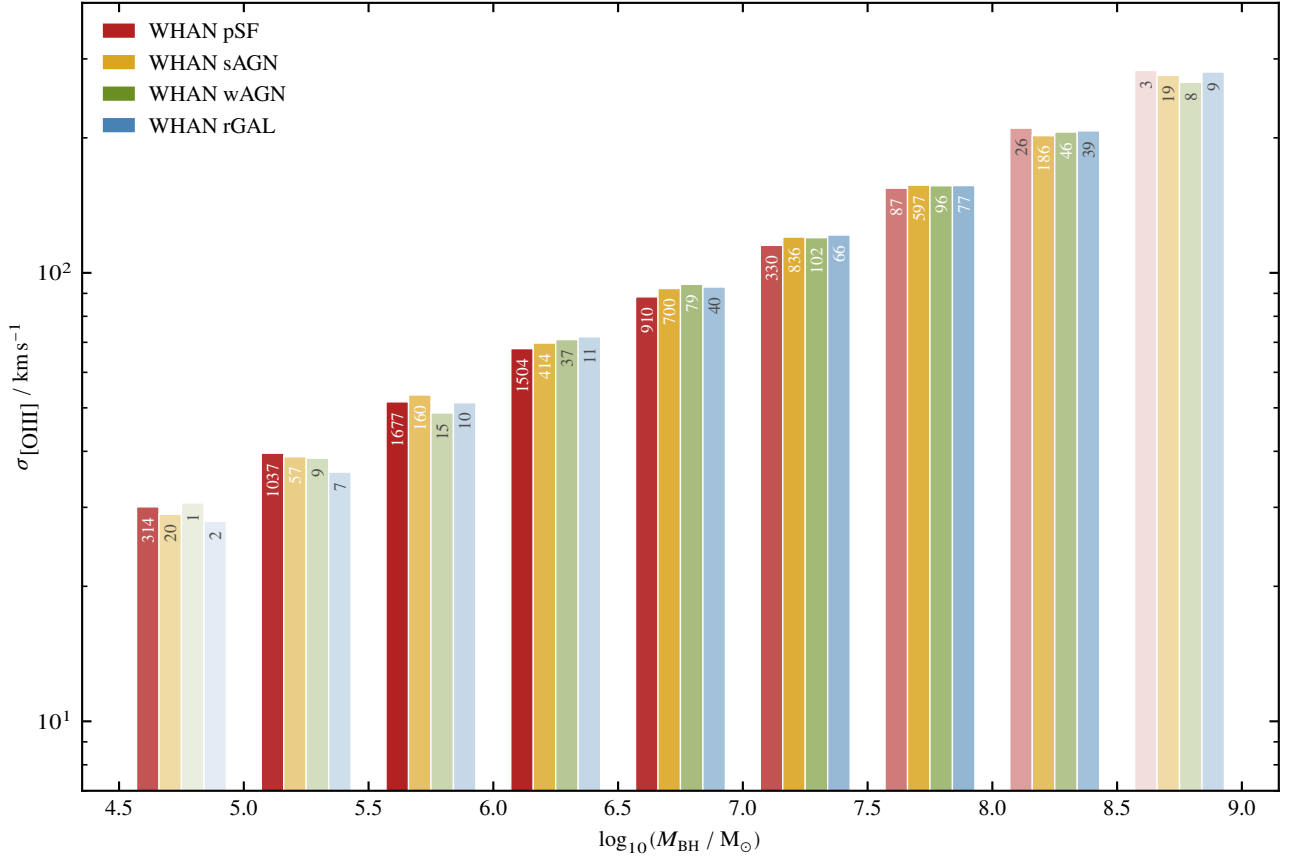


Figure 24

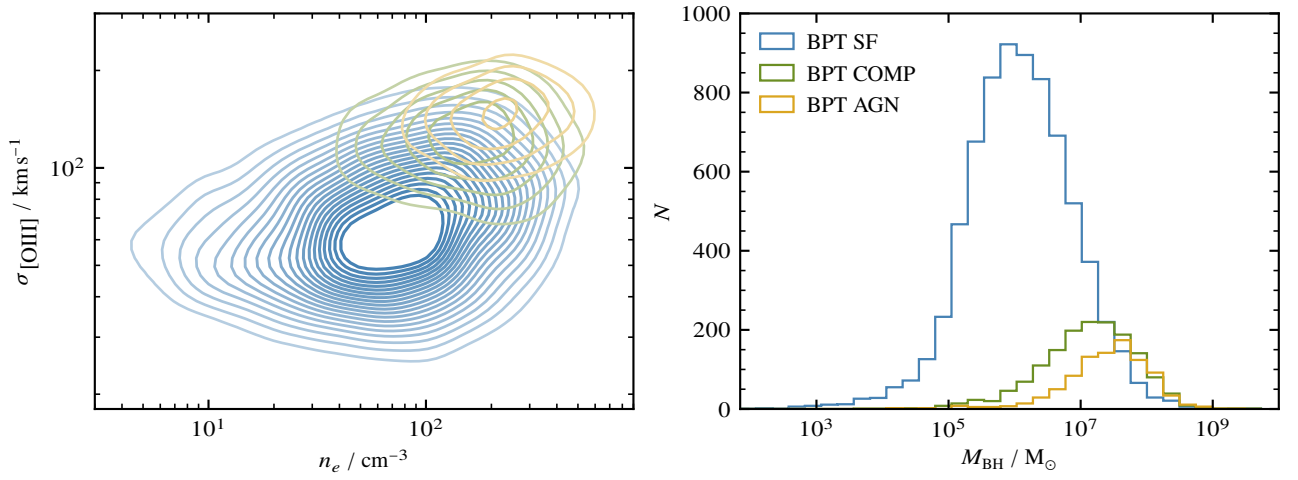


Figure 25

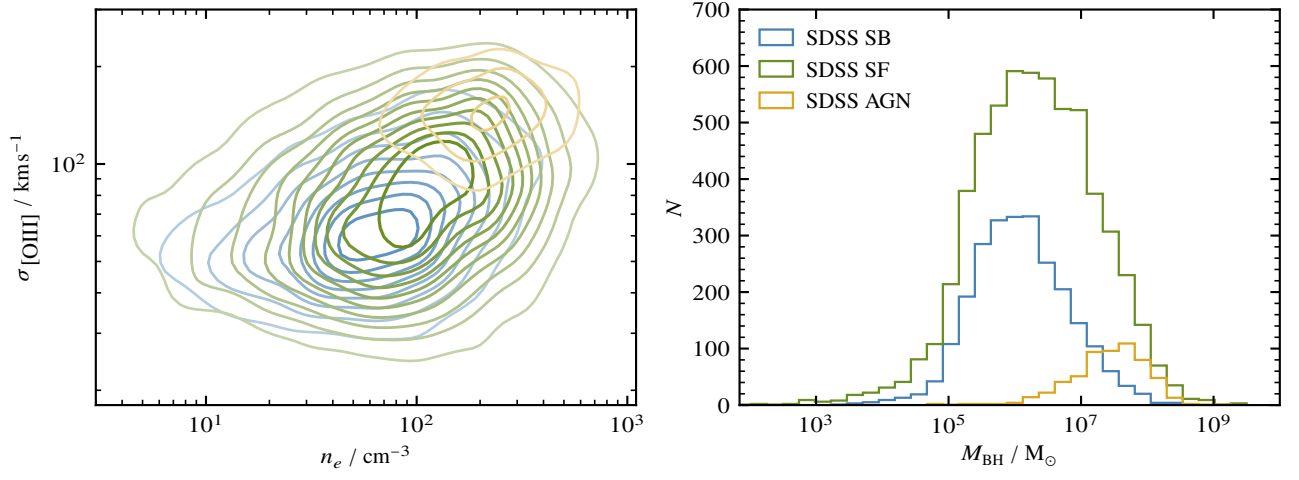


Figure 26

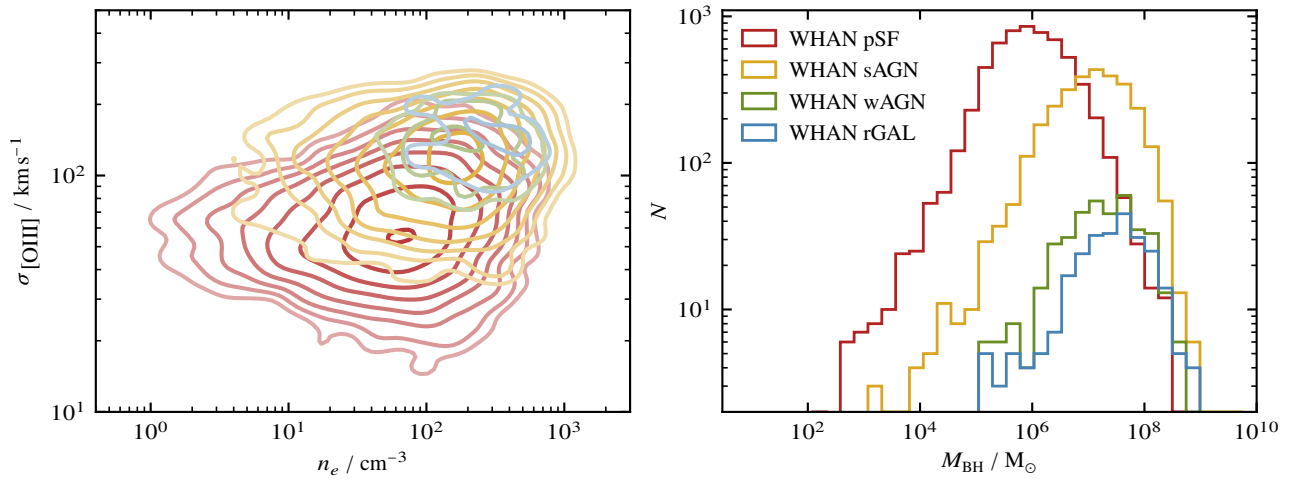


Figure 27

References

- Baldwin, J. A. et al. *Publications of the Astronomical Society of the Pacific* (1981)
Kauffmann, G. et al. *Monthly Notices of the Royal Astronomical Society* (2003)
Kewley, L. J. et al. *The Astrophysical Journal* (2001)
Kollmeier, J. A. et al. *The Astronomical Journal* (2026)
Wright, E. L. et al. *The Astronomical Journal* (2010)

Appendix

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1 SELECT TOP 10000
2     g.sii_6717_flux, g.sii_6731_flux, g.nii_6584_flux, g.oi_6300_flux, g.oiii_5007_flux,
3     g.h_alpha_flux, g.h_beta_flux,
4     g.h_alpha_eqw, g.nii_6584_eqw, g.oiii_sigma,
5     w.w1mpro, w.w2mpro, w.w3mpro,
6     s.z, s.class, s.subclass, s.bestobjid
7 FROM SpecObj AS s
8 JOIN galSpecLine AS g ON s.specobjid = g.specobjid
9 JOIN PhotoTag AS p ON s.bestobjid = p.objid
10 OUTER APPLY ( SELECT TOP 1 x.wise_cntr
11                FROM wise_xmatch AS x
12                WHERE p.objid = x.sdss_objid AND x.match_dist < 3
13                ORDER BY x.match_dist ASC
14            ) AS y
15 LEFT JOIN wise_allsky AS w ON y.wise_cntr = w.cntr
16 WHERE s.class = 'GALAXY'
17 AND s.subclass != 'BROADLINE'
18 AND s.subclass != 'AGN BROADLINE'
19 AND s.subclass != 'STARFORMING BROADLINE'
20 AND s.subclass != 'STARBURST BROADLINE'
21 AND s.snmedian_r > 5.0
22 AND s.z < 0.35
23 AND g.sii_6717_eqw < 0.0
24 AND g.sii_6731_eqw < 0.0
25 AND g.nii_6584_eqw < 0.0
26 AND g.oi_6300_eqw < 0.0
27 AND g.oiii_5007_eqw < 0.0
28 AND g.h_alpha_eqw < 0.0
29 AND g.h_beta_eqw < 0.0
30 AND (2.355 * g.sigma_forbidden) < 500
31 AND (2.355 * g.sigma_balmer) < 500
32 AND g.sii_6717_flux > 3 * g.sii_6717_flux_err
33 AND g.sii_6731_flux > 3 * g.sii_6731_flux_err
34 AND g.nii_6584_flux > 3 * g.nii_6584_flux_err
35 AND g.oi_6300_flux > 3 * g.oi_6300_flux_err
36 AND g.oiii_5007_flux > 3 * g.oiii_5007_flux_err
37 AND g.h_alpha_flux > 3 * g.h_alpha_flux_err
38 AND g.h_beta_flux > 3 * g.h_beta_flux_err
39 ORDER BY s.specobjid ASC

```